Ex:No: 01
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Study Of Various Network Commands Used In Linux And Windows

Aim:

Study of Various Network Commands Used in Linux and Windows.

Basic Commands:

1) arp -a:

ARP is short for address resolution protocal. It will show the ip and MAC address mapping of Connected devices.

OUTPUT:

→ arp -a

Interface: 172.16.10.51 --- 0x4

Internet address	physical Address	Type
172.16.8.1	7c-5a-1c-cf-be-45	dynamic
172.16.8.17	d4-3d-7e-14-85-5b	dynamic
172.16.8.63	f8-bc-12-90-45-97	dynamic
172.16.8.125	00-27-0e-13-f6-0d	dynamic

2) hostname:

It display the name of host machine.

OUTPUT:

→ hostname.

DESKTOP-V8C2B3D.

3) ipconfig /all or ifconfid -a:

It displays detail configuration information about the TCP/IP connected including Router, Gateway, DNS, ethernet.

OUTPUT:

→ ipconfig /all

Window IP Configuration.

Hostname : DESKTOP-V8C2B30

Primary DNS Suffix.

Node Type : Hybrid.

Ip routing Enable : No

WINS proxy Enabled : No.

Ethernet adapter Ethernet 2.

Connection-Specific DNS Suffix:

Description : Intel (R). Ethernet Connection

physical address : D5-BB-C1-C5-CD-76

DHCP Enable : No

IPV4 address : 172.16.10.51

4) nbtstat -a :

Display the net Bios information of the Specified host.

OUTPUT:

→ nbtstat -a 172.16.11.102

Ethernet 2:

Node IP Address [172.16.10.51] ScopeId: []

Net Bios Remote Machine Name Table.

Name	Type	Status.
Desktop-GNI8APV<00>	UNIQUE	Registered.
Work Group	GROUP	Registered
Desktop-GNI8APV<20>	UNIQUE	Registered

Mac Address : 50-9A-9C-34-08-F9.

5) netstat:

display Statics about the TCP/IP connections.

OUTPUT:

→ net stat

Active Connection.

Pocket	Local Address	Foreign Address	State
TCP	178.16.70.31.7650	Desktop (9H713158)	Time_coait
TCP	172.16.10.51.5401	4.233.25.240	Established.

6) nslookup:

Display DNS details of the specified Computer.

OUTPUT:

→ nslookup faiharish.xyz

Server : Unknown

Address : 178-16.8.1

Non authoritive answer:

Name : faiharish.xyz

Address : 216.198.79.1

7) Pathping:

It is a combination of ping and Tracert commands.

OUTPUT:

→ pathping saiharish.xyz

Training route to saiharish.xyz [216.198.29.1]

0 DESKTOP-V812B30 (172.16.10.51)

1 172.16.8.1

2 mail.hatoid.com (83.156.9.89)

3 *

8) Ping:

It is used to test the connectivity between two nodes.

OUTPUT:

→ ping saiharish.xyz

Pinging saiharish.xyz [216.198.79.1] with 32 bytes of dat

Reply from 216.198.79.1 bytes=32 time=245 TTL:2

9) Route:

It prints the routing table of the network.

OUTPUT:

→ route PRINT

IPV4 Route Table

Network Destination	Network Gateway	Interface
0.0.0.0	0.0.0.0 12.16.0.1	172.6.10.31

RESULT:

Thus the basic networking commands have been practiced.